Week 13 Assignment

Mixed-effects models II

Render this file to PDF and submit both the `.qmd` and the rendered `.pdf`, along with your Data Safari submission, via Canvas before Week 14's lecture (Monday). See the syllabus for the late policy.

Note: This week also includes a Data Safari exercise — a separate, short assignment. Both are due at the same time.

```
library(tidyverse)
library(lme4)
library(lmerTest)
library(marginaleffects)
```

Part A: Random slopes. Using `ChickWeight` (diet 1 only):

1. Fit a random-intercept model and a random-slope model for `weight ~ Time`, with `Chick` as the grouping variable.
2. Compare them with `anova()`. Which is better supported?
3. Use `avg_slopes()` on both. What changes between the two, and why?

Part B: Design structure.

4. A researcher counts parasites in 6 fish from each of 10 ponds, in 3 lakes. What is the random-effects structure? Write the `lmer()` formula.
5. Each of 15 leaves is cut into 3 discs, and each disc is assigned to a different light condition. Are leaf and light nested or crossed? Write the formula.